Q] How many rectangles in grid?

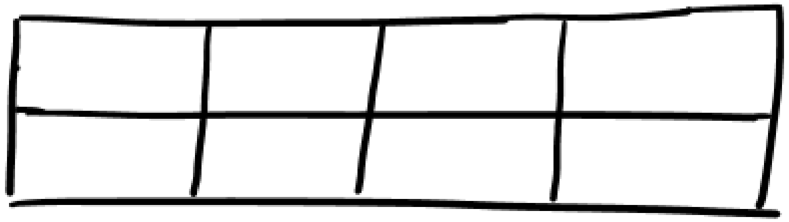
(A) 6

(B) 24

~~(C)~~ 30

(D) N.O.T.A

[GATE 2016]



M1

1 cell = 8

2 cell = horizontal → 6

vertical → 4

3 cell = 4

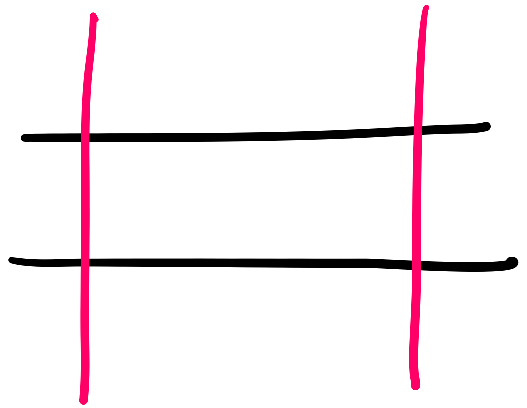
4 cell = 3 + 2 = 5

6 cell = 2

8 cell = 1

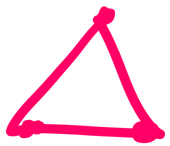
Ans = 8 + 6 + 4 + 4 + 5 + 2 + 1

= 30 //



1 rectangle = 2 Horizontal line &

2 vertical line

1 rectangles = 2 Horizontal line (f)

2 vertical line

$\Rightarrow$ All rectangles = ${}^{n}C_2 \times {}^{n}C_2$

Q] How many rectangles in grid?

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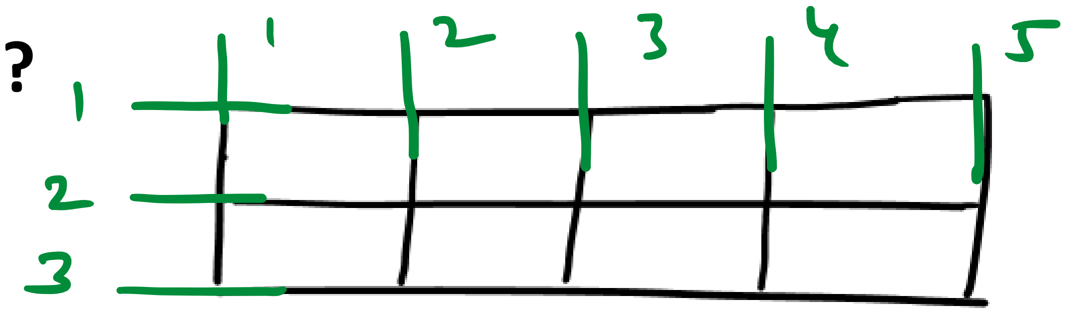
[GATE 2016]

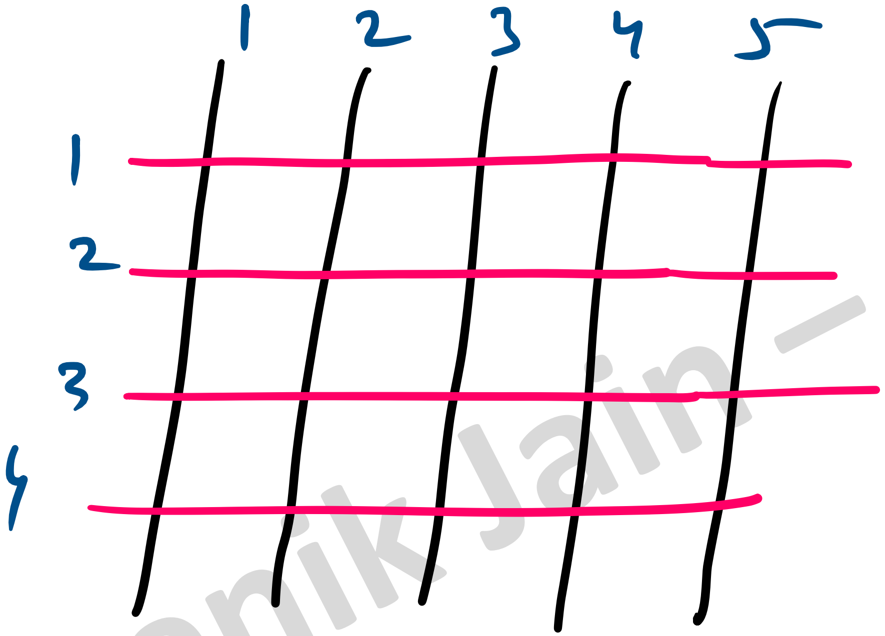


M2

$$ {}^3C_2 \times {}^5C_2 $$

$$ = 3 \times \frac{5 \times \cancel{4}^{\,2}}{\cancel{2} \times 1} = 30 $$

Q] 4 horizontal parallel lines , 5 parallel lines cut each other. How many parallelogram is formed?



$$^5C_2 \times {^4C_2}$$

$$= \frac{5\times\cancel{4}}{\cancel{2}} \times \frac{4\times3}{\cancel{2}}$$

$$= 20\times 3$$

$$= 60$$

Q] Everybody in room shake hands with everybody else. The total number of handshakes is 66. Total number of person in room?

n person $\rightarrow$ n hands = 12 (Ans)

$$^nC_2 = 66$$

$$\frac{n(n-1)}{2} = 66$$

$$n^2 - n = 132 \Rightarrow \overset{12}{n}\,(\overset{11}{n-1}) = 132$$

Q] Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

(A) 20

(B) 10

(C) 8

(D) 5 [GATE 2015]

M1 $\{A, B, C, D, E\}$

M2

Selection

$$\boxed{{}^nC_2}$$

$${}^5C_2 = \frac{5 \cdot 4}{2 \cdot 1} = 10$$

Q] A three member committee has to be formed from a group of 9 people. How many such distinct committee can be formed?

[GATE 2018]

$$^{9}C_{3} = \frac{9\times8\times7}{3\times2\times1}$$

$$= 84$$

Q] The manufacturing area of a plant is divided into four quadrants. Four machines have to be located, one in each quadrant . The total number of possible layouts is

(A) 4

(B) 8

(C) 16

(D) 24 [GATE 1995]

M1

Permutation

$^4P_4 = 4! = 4 \times 3 \times 2 \times 1 = 24$

M2

M_1 M_2 M_3 M_4

$\underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 4! = 24$

Q] The number of ways in which 5 pictures can be hung from 7 picture nails on the wall is

M1

$$^7C_5 \times 5! = \frac{7 \times 6}{2} \times 5! = 2520$$

M2

$$^7P_5 = \frac{7!}{(7-5)!} = \frac{7!}{2!} = 2520$$

Q] In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games both the players were girls and in 190 games both the players were boy. The number of games in which one player was a boy and other was girl is

(Combination)

① Let us say : m Boys

n Girls

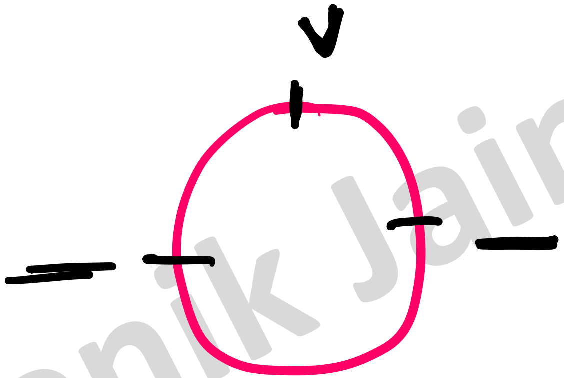
② $^{n}C_2 = 45 \rightarrow \frac{n(n-1)}{2} = 45 \rightarrow n = 10$ //

$^{m}C_2 = 190 \rightarrow \frac{m(m-1)}{2} = 190 \rightarrow m = 20$ //

$$^{20}C_1 \times {}^{10}C_1 = 20 \times 10 = 200$$

V A S

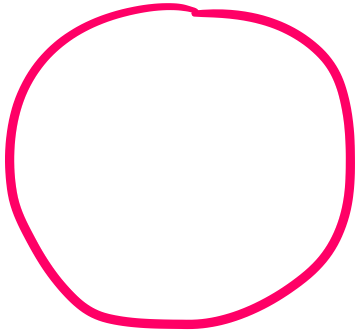
$\underline{3} \times \underline{2} \times \underline{1} = 6$ ways $= 3!$



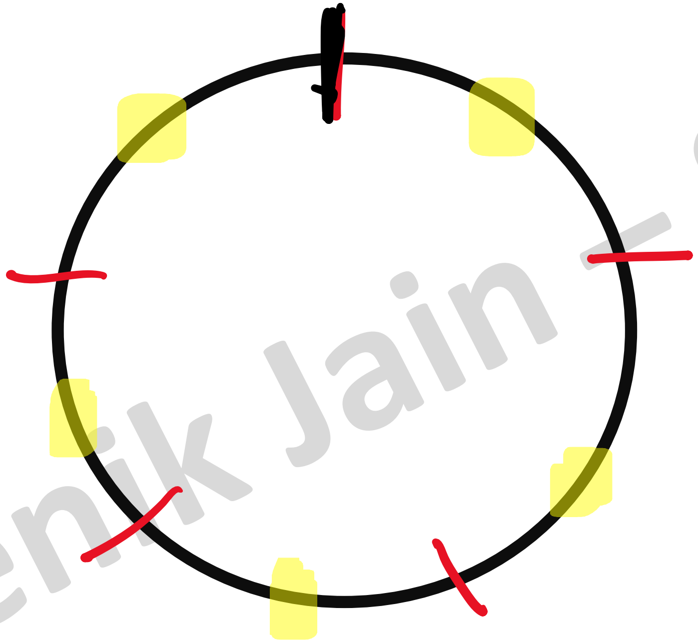
$\rightsquigarrow \underline{2}\ \underline{1} = 2! = 2$

A S

n person ⇒ Circular Table ⇒ No of ways $= (n-1)!$

 $\longrightarrow (n-1)!$

Q] There are 5 gentleman and 4 ladies to dine at round table. In how many ways can they seat themselves so that no two ladies are together?



$4! \times {}^5P_4$

$= 4! \times 5!$

$= 2880$

Q] The sum of all the numbers greater than 10000 formed by using digits 0,2,4,6,8 (no digit repeated in any number) is equal to ?

① Jo chahiye = Total − Jo Nahi chahiye

② Total (Sum → 4 & 5 no's)

$$Sum = (5-1)! \times (0+2+4+6+8) \times (11111)$$
$$= 533280$$

③ Jo Nahi chahiye ⇒ {2, 4, 6, 8}

$$Sum = (4-1)! \times (2+4+6+8) \times (1111)$$

③ To Nahi chahiye $\Rightarrow \{2, 4, 6, 8\}$

$$Sum = (4-1)! \times (2+4+6+8) \times (1111)$$

$$= 133320$$

$$Ans = 5333280 - 133320$$

$$= 5199960$$ //

$$
\begin{array}{r}
1\,0\,0\,0 \\
+\;1\,2\,3\,3 \\
+\;0\,1\,0\,0 \\
+\;0\,0\,1\,5 \\
\hline
2\,3\,4\,8 \\
\hline
\end{array}
$$

$\boxed{\text{LOGIC}}$

$$
\begin{array}{lcr}
1000(1+1+0+0) & \Rightarrow & 2000 \\
 & & + \\
+\;100(0+2+1+0) & \Rightarrow & 300 \\
 & & + \\
+\;10(0+3+0+1) & \Rightarrow & 40 \\
 & & + \\
+\;1(0+3+0+5) & \Rightarrow & 8 \\
\hline
 & & 2348 \\
\hline
\end{array}
$$

$$\begin{array}{r} 1\,1\,1\,1 \\ +\,1\,9\,9\,9 \\ \hline 3\,1\,1\,0 \\ \hline \end{array}$$

$$\cancel{1000(1+1)}^{\,2000} +$$

$$\cancel{100(1+9)}^{\,1000} +$$

$$\cancel{10(1+9)}^{\,100} +$$

$$\cancel{1(1+9)}^{\,10}$$

$$\frac{\qquad\qquad}{\underline{3110}}$$

1, 2, 3, 4 $\Rightarrow$ Sum of all 4 digit no from it

$\boxed{1}$ _ _ _ $\boxed{2}$ _ _ _ $\boxed{3}$ _ _ _ $\boxed{4}$ _ _ _

1 2 3 4	2 1 3 4	3 1 2 4	4 1 2 3
1 2 4 3	2 1 4 3	3 1 4 2	4 1 3 2
1 3 2 4	2 3 1 4	3 2 1 4	4 2 1 3
1 3 4 2	2 3 4 1	3 2 4 1	4 2 3 1
1 4 2 3	2 4 1 3	3 4 1 2	4 3 1 2
1 4 3 2	2 4 3 1	3 4 2 1	4 3 2 1

$$\therefore \text{Sum} = 1000\,(6\times1 + 6\times2 + 6\times3 + 6\times4)$$
$$+\ 100\,(6\times1 + 6\times2 + 6\times3 + 6\times4)$$
$$+\ 10\,(6\times1 + 6\times2 + 6\times3 + 6\times4)$$
$$+\ 1\,(6\times1 + 6\times2 + 6\times3 + 6\times4)$$

$$= 6\,(1+2+3+4)\,(\overbrace{1000+100+10+1}^{1111})$$

$$(4-1)! \times (\text{Sum of all digits}) \times (1111)$$

$$\boxed{\text{Sum of } \underline{n \text{ digits}} \text{ (w/o)} = (n-1)! \times (\text{Sum of all digits}) \times (11111\ldots n \text{ times})}$$

$(4-1)! \times (\text{Sum of all digits}) \times (1111)$

Q] 1, 2, 3, 4 $\Rightarrow$ Sum of all 4 digit no from it

$$\text{Sum} = (4-1)! \times (1+2+3+4) \times (1111)$$

"$\underline{\text{with repetition}}$"

$\{1, 2, 3, 4\}$ ✓

$\boxed{1}$ $\underline{4}$ $\underline{4}$ $\underline{4}$ $= 4^3 = n^{n-1}$

$$\text{Sum of } \underline{n \text{ digits}} \text{ (with repetition)} = n^{n-1} \times (\text{Sum of all digits}) \times (11111\ldots n \text{ times})$$